

DRUG REPURPOSING DEMO

Query : 'hypertension' → hypertension

Model : HGT

Scoring 100 drug candidates ...

Computing disease similarity ...

OUTPUT 1 — TOP 5 CANDIDATE DRUGS

Disease : hypertension | Model : HGT

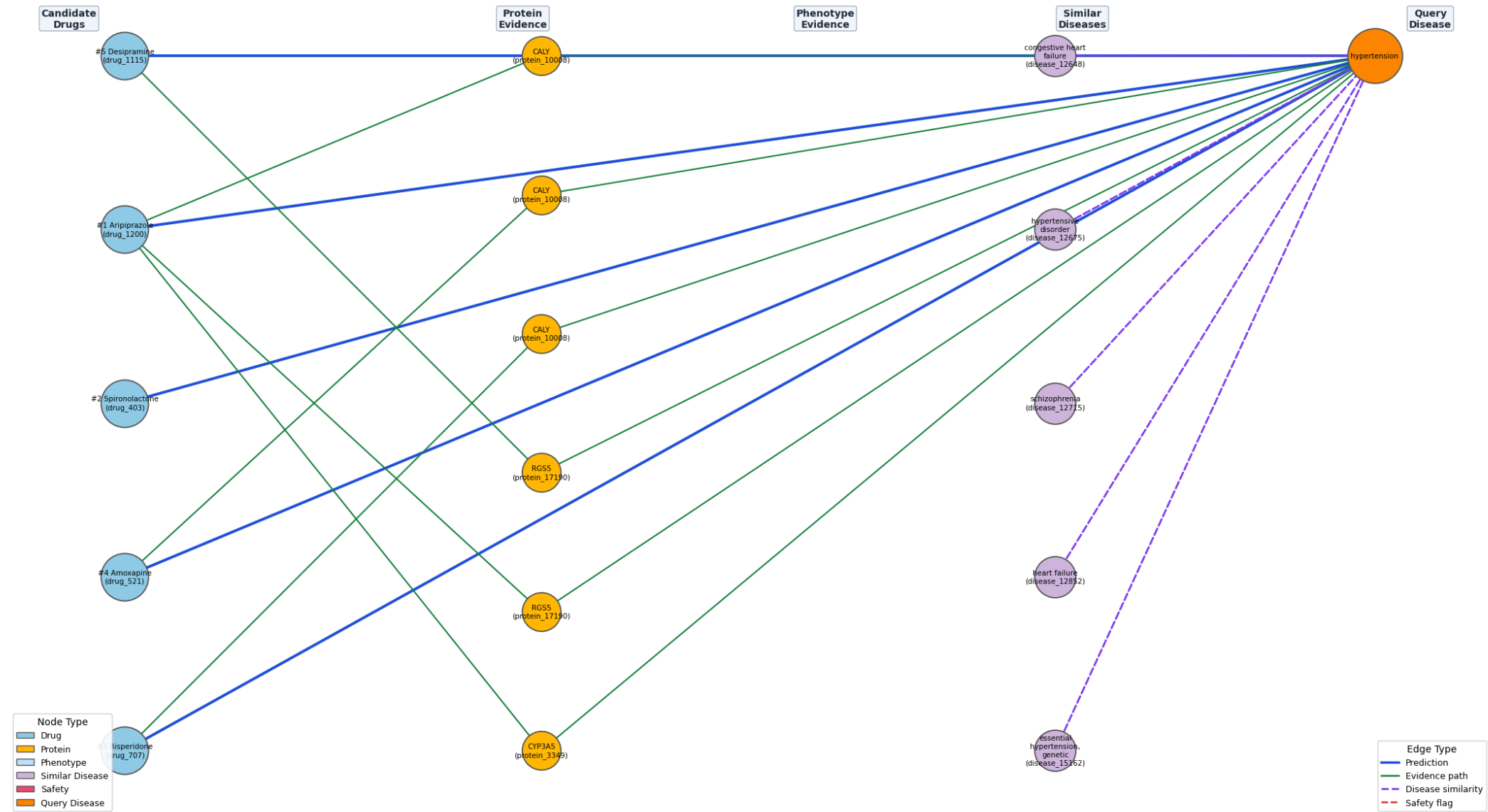
	Drug	Model Score	Sim. Support	Rerank Score	Risk	Evidence	Safety Flags	Known Treatment For
Rank								
1	Aripiprazole (drug_1200)	0.554106	0.587499	0.651633	low	4	0	bipolar disorder, schizophrenia, autism spectrum disorder (+1 more)
2	Spironolactone (drug_403)	0.553825	0.706262	0.521170	low	1	0	Barter disease, primary aldosteronism, congestive heart failure (+1 more)
3	Risperidone (drug_707)	0.554088	0.587499	0.501623	low	1	0	autism susceptibility 1, manic bipolar affective disorder, bipolar disorder (+2 more)
4	Amoxapine (drug_521)	0.554136	0.573907	0.498252	low	1	0	dysthymic disorder, neurotic disorder, anxiety disorder
5	Desipramine (drug_1115)	0.553715	0.573907	0.498020	low	1	0	dysthymic disorder, neurotic disorder, anxiety disorder

OUTPUT 2 — TOP 10 SIMILAR DISEASES

	Similar Disease	Similarity Score	Embedding Cos	Shared Proteins	Shared Effects	Shared Drugs
Rank						
1	hypertensive disorder (disease_12675)	0.799431	0.996750	12	0	51
2	essential hypertension, genetic (disease_15162)	0.733706	0.827450	12	0	2
3	schizophrenia (disease_12715)	0.596113	0.975533	6	0	0
4	congestive heart failure (disease_12648)	0.593885	0.997253	6	0	8
5	heart failure (disease_12852)	0.579487	0.994419	6	0	2
6	anxiety disorder (disease_15492)	0.577918	0.935944	6	0	0
7	kidney disease (disease_12844)	0.571030	0.596280	1	0	0
8	unipolar depression (disease_12863)	0.570691	0.908018	6	0	0
9	bipolar disorder (disease_12626)	0.562907	0.867104	6	0	0
10	dysthymic disorder (disease_2746)	0.554288	0.944373	5	0	0

OUTPUT 3 — EVIDENCE & SIMILARITY SUBGRAPH

Drug Repurposing Evidence Graph — hypertension (model: HGT)
Drugs · Protein Evidence · Phenotypes · Similar Diseases · Safety



Subgraph saved → /mnt/data/demo_subgraph.png

OUTPUT 4 — RAG EXPLANATION & SUMMARY

Disease query : hypertension
Model : HGT

SIMILAR DISEASES USED AS CONTEXT

Diseases most biologically similar to the query were identified using:
embedding cosine similarity · disease ontology edges · shared proteins
PPI-expanded protein bridges · phenotype overlap · shared therapeutics

- hypertensive disorder (disease_12675) (similarity=0.799, embedding=0.997, shared proteins=12, shared effects=0)
- essential hypertension, genetic (disease_15162) (similarity=0.734, embedding=0.827, shared proteins=12, shared effects=0)
- schizophrenia (disease_12715) (similarity=0.596, embedding=0.976, shared proteins=6, shared effects=0)
- congestive heart failure (disease_12648) (similarity=0.594, embedding=0.997, shared proteins=6, shared effects=0)
- heart failure (disease_12852) (similarity=0.579, embedding=0.994, shared proteins=6, shared effects=0)

TOP 5 CANDIDATE DRUGS

Rank 1 – Aripiprazole (drug_1200)

Currently known to treat : bipolar disorder, schizophrenia, autism spectrum disorder (+1 more)
Raw model score : 0.554
Similar disease support : 0.587
Final rerank score : 0.652
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Shared protein target: Aripiprazole (drug_1200) --[drug_enzyme_protein]--> CYP3A5 (protein_3349) <--[disease_associated_with_protein]-- hypertension (disease_1569)
- Protein interaction bridge: Aripiprazole (drug_1200) --[drug_enzyme_protein]--> CYP3A7 (protein_3316) --[protein_interacts_with_protein]--> CYP3A5 (protein_3349) <--[disease_associated_with_protein]-- hypertension (disease_1569)
- Protein interaction bridge: Aripiprazole (drug_1200) --[drug_target_protein]--> ADRB2 (protein_3281) --[protein_interacts_with_protein]--> RGS5 (protein_17190) <--[disease_associated_with_protein]-- schizophrenia (disease_12715)
- Protein interaction bridge: Aripiprazole (drug_1200) --[drug_target_protein]--> DRD1 (protein_3816) --[protein_interacts_with_protein]--> CALY (protein_10008) <--[disease_associated_with_protein]-- heart failure (disease_12852)

Evidence profile:

- Shared protein bridges : 1
- Protein interaction bridges : 3
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

schizophrenia (disease_12715) | bipolar disorder (disease_12626)

Rank 2 – Spironolactone (drug_403)

Currently known to treat : Bartter disease, primary aldosteronism, congestive heart failure (+1 more)
Raw model score : 0.554
Similar disease support : 0.706
Final rerank score : 0.521
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Related disease transfer: Spironolactone (drug_403) --[known treatment]--> hypertensive disorder (disease_12675) --[disease_parent_child_disease]--> hypertension (disease_1569)

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 0
- Disease-to-disease transfer : 1
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:
hypertensive disorder (disease_12675) | congestive heart failure (disease_12648)

Rank 3 – Risperidone (drug_707)

Currently known to treat : autism susceptibility 1, manic bipolar affective disorder, bipolar disorder (+2 more)
Raw model score : 0.554
Similar disease support : 0.587
Final rerank score : 0.502
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:
• Protein interaction bridge: Risperidone (drug_707) --[drug_target_protein]--> DRD1 (protein_3816) --[protein_interacts_with_protein]--> CALY (protein_10008) <--[disease_associated_with_protein]

Evidence profile:
• Shared protein bridges : 0
• Protein interaction bridges : 1
• Disease-to-disease transfer : 0
• Phenotype / effect overlap : 0

Safety signals:
• No direct contraindication or adverse overlap signal found.

Support from similar diseases:
schizophrenia (disease_12715) | bipolar disorder (disease_12626)

Rank 4 – Amoxapine (drug_521)

Currently known to treat : dysthymic disorder, neurotic disorder, anxiety disorder
Raw model score : 0.554
Similar disease support : 0.574
Final rerank score : 0.498
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:
• Protein interaction bridge: Amoxapine (drug_521) --[drug_target_protein]--> DRD1 (protein_3816) --[protein_interacts_with_protein]--> CALY (protein_10008) <--[disease_associated_with_protein]

Evidence profile:
• Shared protein bridges : 0
• Protein interaction bridges : 1
• Disease-to-disease transfer : 0
• Phenotype / effect overlap : 0

Safety signals:
• No direct contraindication or adverse overlap signal found.

Support from similar diseases:
anxiety disorder (disease_15492) | dysthymic disorder (disease_2746)

Rank 5 – Desipramine (drug_1115)

Currently known to treat : dysthymic disorder, neurotic disorder, anxiety disorder
Raw model score : 0.554
Similar disease support : 0.574
Final rerank score : 0.498
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:
• Protein interaction bridge: Desipramine (drug_1115) --[drug_target_protein]--> ADRB2 (protein_3281) --[protein_interacts_with_protein]--> RGS5 (protein_17190) <--[disease_associated_with_protein]

Evidence profile:
• Shared protein bridges : 0

- Protein interaction bridges : 1
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

anxiety disorder (disease_15492) | dysthymic disorder (disease_2746)

INTERPRETATION NOTE

These are graph-based research hypotheses, not clinical prescriptions. The strongest candidates combine: high model score + strong mechanistic evidence (especially shared protein bridges) + support from biologically similar diseases + no direct contraindication signals.

Always validate predictions against current clinical literature before drawing any experimental or clinical conclusions.

Structured RAG report table:

rank	drug	known_treatment_for	model_score	similar_disease_support	rerank_score	risk	evidence_summary	safety_summary
1	Aripiprazole (drug_1200)	bipolar disorder, schizophrenia, autism spectrum disorder (+1 more)	0.554106	0.587499	0.651633	low	Shared protein target: Aripiprazole (drug_1200) -- [drug_enzyme_protein]--> CYP3A5 (protein_3349) <--[disease_associa...	No direct contraindication or adverse overlap signal found.
2	Spironolactone (drug_403)	Bartter disease, primary aldosteronism, congestive heart failure (+1 more)	0.553825	0.706262	0.521170	low	Related disease transfer: Spironolactone (drug_403) --[known treatment]--> hypertensive disorder (disease_12675) --[...	No direct contraindication or adverse overlap signal found.
3	Risperidone (drug_707)	autism susceptibility 1, manic bipolar affective disorder, bipolar disorder (+2 more)	0.554088	0.587499	0.501623	low	Protein interaction bridge: Risperidone (drug_707) -- [drug_target_protein]--> DRD1 (protein_3816) -- [protein_interac...	No direct contraindication or adverse overlap signal found.
4	Amoxapine (drug_521)	dysthymic disorder, neurotic disorder, anxiety disorder	0.554136	0.573907	0.498252	low	Protein interaction bridge: Amoxapine (drug_521) -- [drug_target_protein]--> DRD1 (protein_3816) -- [protein_interacts...	No direct contraindication or adverse overlap signal found.
5	Desipramine (drug_1115)	dysthymic disorder, neurotic disorder, anxiety disorder	0.553715	0.573907	0.498020	low	Protein interaction bridge: Desipramine (drug_1115) -- [drug_target_protein]--> ADRB2 (protein_3281) -- [protein_inter...	No direct contraindication or adverse overlap signal found.